

APM 115: Mathematical Modeling
Ivy League Inbreeding - Markov Processes

In an attempt to bridge the gap between students attending Harvard and those attending Yale, a specially appointed intramural committee proposed a yearly social event, where students from both universities would get to meet each other and interact. This suggestion was applauded on both sides of the Charles, until a statistician and a geneticist from MIT coauthored a paper predicting the demise of one of these great institutions within less than ten generations. The assumptions behind their argument were that this social event would inevitably lead to romantic relationships, some of which would eventually yield offspring (typically two children and a dog), and that the children will end up going to either Harvard or Yale. You have been asked to reexamine the MIT findings and report to the boards of trustees about the advisability of this social event. Here's what you need to know:

Two genes govern a person's tendency to attend a university, the dominant h gene and the recessive y gene. A person with two y genes (defined Y) will attend Yale, while someone with two h genes (defined H) or a combination of h and y (defined M for mixed) will attend Harvard.

For simplicity, we shall assume that two people will marry and have children only if their respective parents have the same genotype. For example, a person with parents with the HM genotype (one parent is hh and the other is hy) will only marry a person whose parents are also of the HM genotype. Mathematically, this assumption gives the same result as assuming a couple's children marrying each other, which is perhaps easier to understand, but of course has very different social implications... (Without such an assumption, the system will become difficult to track as a Markov chain. It would be useful to ponder over this for a moment.)

Model this system as a Markov process:

1. Define the states and construct the transition matrix of the system, in canonical form.
2. Numerically evolve the probability vector (i.e. multiply the probability vector by the transition matrix) to equilibrium for some different initial conditions as prescribed below.

How does the initial size of the student populations affect the final probabilities? Try:

- i. equal number of people of a certain phenotype: $Y = 1/2$, $H = 1/4$, $M = 1/4$
- ii. equal number of people of a certain genotype: $Y = 1/3$, $H = 1/3$, $M = 1/3$

Do you agree with the MIT scientists?

We shall assume that the first-generation students are more open-minded and marry within the population randomly. This can give you the initial condition for the genotypes of the couples. After the initial time, the subsequent generations develop this prejudice and only marry people with parents that are of the same genotype.

3. Use the fundamental matrix and related analysis approach covered in class to analyze this problem. For example, calculate the number of generations to absorption, as well as the probability of each outcome as function of the initial state.

4. Compare your results from 2i and 2ii to a stochastic individual agent simulation: begin with initial set of agents with phenotype or genotype distributions as in 2i and 2ii and keep track of the evolving populations based on the imposed rules. Use a random number generator to decide on probabilistic outcome when needed. Perform many simulations and compare some key statistics to those found from the fundamental matrix.